Evidence to Proof Route Onramp

Expert inference validation report

Focus question
How did source evidence become a falsification-aware proof route?

Run Id
fixture.evidence_to_proof_route_onramp.base

Target
p-vs-np

Trace Id
trace.c723abc8ccf1edcc

Workflow Id
evidence_to_proof_route_onramp

Workflow Name
Evidence to Proof Route Onramp

Workflow Version
1.0.0

Canonical packet SHA-256
059a8ac50e1dcf93eff97b3e5de4e4488bde25f27ad54c5690c7f303e00b6f10

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1 Expert reading contract

Public rationale boundary

Inspectable public rationale and evidence only. The inventory must not contain private chain-of-thought or derive scientific claims from layout, chronology, control authorization, or artifact integrity.

This document records inspectable premises, public warrants, evidence, conclusions, uncertainty, scope, authority, failures, and obligations. It does not expose private chain-of-thought. Execution authorization, artifact integrity, scientific epistemic status, and workflow disposition are independent. Chronological precedence does not mean logical or evidential support.

1.1 Status legend

Artifact integrity: whether referenced bytes and receipts validate **Scientific epistemic status:** the bounded support assigned to a claim **Workflow disposition:** what the workflow did with the item

2 Validation dashboard

Control coverage

1.0

Semantic coverage

1.0

Evidence resolution

1.0

Visible trace gaps

0

Required expert contrasts

["source provenance versus normalization", "hub bias versus robust route", "support versus regime-limiting or refuting evidence"]

3 Declared route and topology

Route spine

Provenance to expert route

Topology mode

linear

Step	Skill	Mode / observed state	Consumes	Produces
ingest	latex-kb-ingest	required / observed	["SourceBundle"]	["Ingested KnowledgeBase"]
parse_pvsnp	pvsnp-kb-parser	conditional / observed	["Ingested KnowledgeBase"]	["ParsedKnowledgeBase"]
normalize_theorems	theorem-body-normalizer	required / observed	["Ingested KnowledgeBase", "ParsedKnowledgeBase"]	["NormalizedTheoremCorpus"]
ground_conjecture	kb-main-conjecture-grounder	required / observed	["NormalizedTheoremCorpus"]	["Grounded Conjecture"]

Step	Skill	Mode / observed state	Consumes	Produces
narrate_literature_evolution	target-literature-evolution-narrator	conditional / observed	["Grounded Conjecture", "SourceBundle"]	["TargetLiteratureNarrative"]
connect_proof	kb-proof-connector	required / observed	["Grounded Conjecture"]	["ProofConnectionGraph"]
evaluate_hypotheses	kb-hypothesis-evaluator	conditional / observed	["ProofConnectionGraph"]	["HypothesisAssessment"]
audit_hubs	hub-bias-auditor	required / observed	["ProofConnectionGraph"]	["HubBiasAudit"]
find_entrypoint	kb-entrypoint-finder	required / observed	["ProofConnectionGraph", "HubBiasAudit"]	["EntrypointCandidates"]
pathfind	kb-entrypoint-pathfinder	required / observed	["EntrypointCandidates"]	["ProofRouteCandidates"]
audit_regime_shift	regime-shift-auditor	conditional / observed	["ProofRouteCandidates"]	["RegimeShiftAudit"]
hunt_counterexamples	counterexample-hunter	required / observed	["ProofRouteCandidates"]	["FalsificationPacket"]
expert_handoff	expert-handoff	required / observed	["ProofRouteCandidates", "FalsificationPacket"]	["ExpertRouteDossier"]

4 Profile topology and workflow-specific contrasts

Projection lens

evidence_to_proof_route_onramp

Topology mode

linear

Required expert contrasts

["source provenance versus normalization", "hub bias versus robust route", "support versus regime-limiting or refuting evidence"]

Declared profile topology

No specialized topology applies

Observed specialist roles

No specialist fan-out applies

A specialist fork or join is a workflow relation, not proof that specialist conclusions are mutually consistent. Finite, endpoint, asymptotic, and continuum boundaries remain separate unless an explicit validated inference connects them.

5 Expert inference dossiers

Each dossier is bound to its semantic event ID. Control-only events appear in the lifecycle appendix and are not scientific claims.

5.1 Inference 1: evidence_to_proof_route_onramp.base.ingest.inference

Trace disposition

Complete public dossier

Step / skill

ingest / latex-kb-ingest

Question

At ingest, how does SourceBundle bear on IngestedKnowledgeBase? Which source-grounded route survives bias, regime, and counterexample audits?

Typed premises

[{"statement": "SourceBundle: The source bundle licenses a finite MCSP reduction claim.", "type": "artifact" }]

Transformation / operator

normalize_with_provenance

Bounded public warrant

At ingest, normalize_with_provenance checks SourceBundle before emitting IngestedKnowledgeBase. Authority boundary: May recommend an evidence-bound route; may not assert a theorem or erase provenance and assertion-license limits.

Conclusion

ingest produces IngestedKnowledgeBase: Retain the local-reduction route only in the declared finite regime.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}

Causal parents

["evidence_to_proof_route_onramp.base.ingest.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.b26c5b14983132

Alternative / falsifier / failure links

alternative.fb02202745a464, falsifier.684594eac0ca2b

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["ingest-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.2 Inference 2: evidence_to_proof_route_onramp.base.parse_pvsnp.inference

Trace disposition

Complete public dossier

Step / skill

parse_pvsnp / pvsnp-kb-parser

Question

At parse_pvsnp, how does IngestedKnowledgeBase bear on ParsedKnowledgeBase? Which source-grounded route survives bias, regime, and counterexample audits?

Typed premises

[{"statement": "IngestedKnowledgeBase: The source bundle licenses a finite MCSP reduction claim.", "type": "artifact"}]

Transformation / operator

normalize_with_provenance

Bounded public warrant

At parse_pvsnp, normalize_with_provenance checks IngestedKnowledgeBase before emitting ParsedKnowledgeBase. Authority boundary: May recommend an evidence-bound route; may not assert a theorem or erase provenance and assertion-license limits.

Conclusion

parse_pvsnp produces ParsedKnowledgeBase: Retain the local-reduction route only in the declared finite regime.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}

Causal parents

["evidence_to_proof_route_onramp.base.parse_pvsnp.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.3232ddcaf03acc

Alternative / falsifier / failure links

alternative.3e60b50d8c871c, falsifier.7a94f0b42655d6

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["parse_pvsnp-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.3 Inference 3: evidence_to_proof_route_onramp.base.normalize_theorems.inference

---

Trace disposition

Complete public dossier

Step / skill

normalize_theorems / theorem-body-normalizer

Question

At normalize_theorems, how does IngestedKnowledgeBase bear on NormalizedTheoremCorpus? Which source-grounded route survives bias, regime, and counterexample audits?

Typed premises

[{"statement": "IngestedKnowledgeBase: The source bundle licenses a finite MCSP reduction claim.", "type": "artifact"}]

Transformation / operator

normalize_with_provenance

Bounded public warrant

At normalize_theorems, normalize_with_provenance checks IngestedKnowledgeBase before emitting NormalizedTheoremCorpus. Authority boundary: May recommend an evidence-bound route; may not assert a theorem or erase provenance and assertion-license limits.

Conclusion

normalize_theorems produces NormalizedTheoremCorpus: Retain the local-reduction route only in the declared finite regime.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}

Causal parents

["evidence_to_proof_route_onramp.base.normalize_theorems.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.c6621527158116

Alternative / falsifier / failure links

alternative.f37bf7b158a7ba, falsifier.f6e0eb836f4b00

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["normalize_theorems-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.4 Inference 4: evidence_to_proof_route_onramp.base.ground_conjecture.inference

---

Trace disposition

Complete public dossier

Step / skill

ground_conjecture / kb-main-conjecture-grounder

Question

At ground_conjecture, how does NormalizedTheoremCorpus bear on GroundedConjecture? Which source-grounded route survives bias, regime, and counterexample audits?

Typed premises

[{"statement": "NormalizedTheoremCorpus: The source bundle licenses a finite MCSP reduction claim.", "type": "artifact"}]

Transformation / operator

ground_claim_to_target

Bounded public warrant

At ground_conjecture, ground_claim_to_target checks NormalizedTheoremCorpus before emitting GroundedConjecture. Authority boundary: May recommend an evidence-bound route; may not assert a theorem or erase provenance and assertion-license limits.

Conclusion

ground_conjecture produces GroundedConjecture: Retain the local-reduction route only in the declared finite regime.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}

Causal parents

["evidence_to_proof_route_onramp.base.ground_conjecture.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.ff0cd87caa476a

Alternative / falsifier / failure links

alternative.cc7d723ac890ce, falsifier.63182453b973af

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["ground_conjecture-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.5 Inference 5: evidence_to_proof_route_onramp.base.narrate_literature_evolution.inference

---

Trace disposition

Complete public dossier

Step / skill

narrate_literature_evolution / target-literature-evolution-narrator

Question

At narrate_literature_evolution, how does GroundedConjecture bear on TargetLiteratureNarrative? Which source-grounded route survives bias, regime, and counterexample audits?

Typed premises

[{"statement": "GroundedConjecture: The source bundle licenses a finite MCSP reduction claim.", "type": "artifact"}]

Transformation / operator

assimilate_prior_evidence

Bounded public warrant

At narrate_literature_evolution, assimilate_prior_evidence checks GroundedConjecture before emitting TargetLiteratureNarrative. Authority boundary: May recommend an evidence-bound route; may not assert a theorem or erase provenance and assertion-license limits.

Conclusion

narrate_literature_evolution produces TargetLiteratureNarrative: Retain the local-reduction route only in the declared finite regime.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}

Causal parents

["evidence_to_proof_route_onramp.base.narrate_literature_evolution.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.6d9bdc9d2ecdd2

Alternative / falsifier / failure links

alternative.29ee3c1755a670, falsifier.c55b9a544fe38d

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["narrate_literature_evolution-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.6 Inference 6: evidence_to_proof_route_onramp.base.connect_proof. inference

---

Trace disposition

Complete public dossier

Step / skill

connect_proof / kb-proof-connector

Question

At connect_proof, how does GroundedConjecture bear on ProofConnectionGraph? Which source-grounded route survives bias, regime, and counterexample audits?

Typed premises

[{"statement": "GroundedConjecture: The source bundle licenses a finite MCSP reduction claim.", "type": "artifact"}]

Transformation / operator

construct_and_rank_route

Bounded public warrant

At connect_proof, construct_and_rank_route checks GroundedConjecture before emitting ProofConnectionGraph. Authority boundary: May recommend an evidence-bound route; may not assert a theorem or erase provenance and assertion-license limits.

Conclusion

connect_proof produces ProofConnectionGraph: Retain the local-reduction route only in the declared finite regime.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}

Causal parents

["evidence_to_proof_route_onramp.base.connect_proof.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.a5cf3aa8c367cd

Alternative / falsifier / failure links

alternative.3e879784edd187, falsifier.e7f4f7e7488b21

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["connect_proof-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.7 Inference 7: evidence_to_proof_route_onramp.base.audit_hubs.inference

---

Trace disposition

Complete public dossier

Step / skill

audit_hubs / hub-bias-auditor

Question

At audit_hubs, how does ProofConnectionGraph bear on HubBiasAudit? Which source-grounded route survives bias, regime, and counterexample audits?

Typed premises

[{"statement": "ProofConnectionGraph: The source bundle licenses a finite MCSP reduction claim.", "type": "artifact"}]

Transformation / operator

compare_pre_post_bias_correction

Bounded public warrant

At audit_hubs, compare_pre_post_bias_correction checks ProofConnectionGraph before emitting HubBiasAudit. Authority boundary: May recommend an evidence-bound route; may not assert a theorem or erase provenance and assertion-license limits.

Conclusion

audit_hubs produces HubBiasAudit: Retain the local-reduction route only in the declared finite regime.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}

Causal parents

["evidence_to_proof_route_onramp.base.audit_hubs.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.a1e8d5b2868a2b

Alternative / falsifier / failure links

alternative.7c0ea84c932fdb, falsifier.a3dd870b693e8f

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["audit_hubs-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.8 Inference 8: evidence_to_proof_route_onramp.base.find_entrypoint.inference

---

Trace disposition

Complete public dossier

Step / skill

find_entrypoint / kb-entrypoint-finder

Question

At find_entrypoint, how does ProofConnectionGraph bear on EntrypointCandidates? Which source-grounded route survives bias, regime, and counterexample audits?

Typed premises

[{"statement": "ProofConnectionGraph: The source bundle licenses a finite MCSP reduction claim.", "type": "artifact"}]

Transformation / operator

construct_and_rank_route

Bounded public warrant

At find_entrypoint, construct_and_rank_route checks ProofConnectionGraph before emitting Entrypoint Candidates. Authority boundary: May recommend an evidence-bound route; may not assert a theorem or erase provenance and assertion-license limits.

Conclusion

find_entrypoint produces EntrypointCandidates: Retain the local-reduction route only in the declared finite regime.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}

Causal parents

["evidence_to_proof_route_onramp.base.find_entrypoint.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.cf493724160827

Alternative / falsifier / failure links

alternative.549c52fdf9427f, falsifier.8a984ea6224bf6

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["find_entrypoint-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.9 Inference 9: evidence_to_proof_route_onramp.base.pathfind.inference

---

Trace disposition

Complete public dossier

Step / skill

pathfind / kb-entrypoint-pathfinder

Question

At pathfind, how does EntrypointCandidates bear on ProofRouteCandidates? Which source-grounded route survives bias, regime, and counterexample audits?

Typed premises

[{"statement": "EntrypointCandidates: The source bundle licenses a finite MCSP reduction claim.", "type": "artifact"}]

Transformation / operator

construct_and_rank_route

Bounded public warrant

At pathfind, construct_and_rank_route checks EntrypointCandidates before emitting ProofRouteCandidates. Authority boundary: May recommend an evidence-bound route; may not assert a theorem or erase provenance and assertion-license limits.

Conclusion

pathfind produces ProofRouteCandidates: Retain the local-reduction route only in the declared finite regime.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}

Causal parents

["evidence_to_proof_route_onramp.base.pathfind.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.acd4492512064a

Alternative / falsifier / failure links

alternative.718960964b4131, falsifier.c826e7307f06d9

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["pathfind-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.10 Inference 10: evidence_to_proof_route_onramp.base.audit_regime_shift.inference

Trace disposition

Complete public dossier

Step / skill

audit_regime_shift / regime-shift-auditor

Question

At audit_regime_shift, how does ProofRouteCandidates bear on RegimeShiftAudit? Which source-grounded route survives bias, regime, and counterexample audits?

Typed premises

[{"statement": "ProofRouteCandidates: The source bundle licenses a finite MCSP reduction claim.", "type": "artifact"}]

Transformation / operator

artifact_grounded_transformation

Bounded public warrant

At audit_regime_shift, artifact_grounded_transformation checks ProofRouteCandidates before emitting RegimeShiftAudit. Authority boundary: May recommend an evidence-bound route; may not assert a theorem or erase provenance and assertion-license limits.

Conclusion

audit_regime_shift produces RegimeShiftAudit: Retain the local-reduction route only in the declared finite regime.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}

Causal parents

["evidence_to_proof_route_onramp.base.audit_regime_shift.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.dff0e8ffaf0efe

Alternative / falsifier / failure links

alternative.2e09561a019cad, falsifier.d38138a2cfdcaf

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["audit_regime_shift-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.11 Inference 11: evidence_to_proof_route_onramp.base.hunt_counterexamples.inference

---

Trace disposition

Complete public dossier

Step / skill

hunt_counterexamples / counterexample-hunter

Question

At hunt_counterexamples, how does ProofRouteCandidates bear on FalsificationPacket? Which source-grounded route survives bias, regime, and counterexample audits?

Typed premises

[{"statement": "ProofRouteCandidates: The source bundle licenses a finite MCSP reduction claim.", "type": "artifact"}]

Transformation / operator

adversarially_test_or_split

Bounded public warrant

At hunt_counterexamples, adversarially_test_or_split checks ProofRouteCandidates before emitting FalsificationPacket. Authority boundary: May recommend an evidence-bound route; may not assert a theorem or erase provenance and assertion-license limits.

Conclusion

hunt_counterexamples produces FalsificationPacket: Retain the local-reduction route only in the declared finite regime.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}

Causal parents

["evidence_to_proof_route_onramp.base.hunt_counterexamples.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.a069114828c61a

Alternative / falsifier / failure links

alternative.5edb2627508cba, falsifier.3fa965347548d8

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["hunt_counterexamples-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.12 Inference 12: evidence_to_proof_route_onramp.base.expert_handoff.inference

---

Trace disposition

Complete public dossier

Step / skill

expert_handoff / expert-handoff

Question

At expert_handoff, how does ProofRouteCandidates bear on ExpertRouteDossier? Which source-grounded route survives bias, regime, and counterexample audits?

Typed premises

[{"statement": "ProofRouteCandidates: The source bundle licenses a finite MCSP reduction claim.", "type": "artifact"}]

Transformation / operator

artifact_grounded_transformation

Bounded public warrant

At expert_handoff, artifact_grounded_transformation checks ProofRouteCandidates before emitting ExpertRouteDossier. Authority boundary: May recommend an evidence-bound route; may not assert a theorem or erase provenance and assertion-license limits.

Conclusion

expert_handoff produces ExpertRouteDossier: Retain the local-reduction route only in the declared finite regime.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}

Causal parents

["evidence_to_proof_route_onramp.base.expert_handoff.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.099972813773cf

Alternative / falsifier / failure links

alternative.0a4d095e562c3e, falsifier.9f065afc3ae566

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["expert_handoff-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

---

6 Alternatives, falsifiers, and failed paths

ID	Kind	Inference	Description	Disposition
alternative.fb02202745a464	alternative	evidence_to_proof_route_onramp.base.ingest.inference	Do not emit IngestedKnowledgeBase: Demote the unrestricted reduction route because its assertion license does not cross the model boundary.	rejected
falsifier.684594eac0ca2b	falsifier	evidence_to_proof_route_onramp.base.ingest.inference	A counterexample to the gadget preservation lemma closes the route.	Not recorded
alternative.3e60b50d8c871c	alternative	evidence_to_proof_route_onramp.base.parse_pvsnp.inference	Do not emit ParsedKnowledgeBase: Demote the unrestricted reduction route because its assertion license does not cross the model boundary.	rejected
falsifier.7a94f0b42655d6	falsifier	evidence_to_proof_route_onramp.base.parse_pvsnp.inference	A counterexample to the gadget preservation lemma closes the route.	Not recorded
alternative.f37bf7b158a7ba	alternative	evidence_to_proof_route_onramp.base.normalize_theorems.inference	Do not emit NormalizedTheoremCorpus: Demote the unrestricted reduction route because its assertion license does not cross the model boundary.	rejected

ID	Kind	Inference	Description	Disposition
falsifier.f6e0eb836f4b00	falsifier	evidence_to_proof_route_onramp.base.normalize_theorems.inference	A counterexample to the gadget preservation lemma closes the route.	Not recorded
alternative.cc7d723ac890ce	alternative	evidence_to_proof_route_onramp.base.ground_conjecture.inference	Do not emit GroundedConjecture: Demote the unrestricted reduction route because its assertion license does not cross the model boundary.	rejected
falsifier.63182453b973af	falsifier	evidence_to_proof_route_onramp.base.ground_conjecture.inference	A counterexample to the gadget preservation lemma closes the route.	Not recorded
alternative.29ee3c1755a670	alternative	evidence_to_proof_route_onramp.base.narrate_literature_evolution.inference	Do not emit TargetLiteratureNarrative: Demote the unrestricted reduction route because its assertion license does not cross the model boundary.	rejected
falsifier.c55b9a544fe38d	falsifier	evidence_to_proof_route_onramp.base.narrate_literature_evolution.inference	A counterexample to the gadget preservation lemma closes the route.	Not recorded
alternative.3e879784edd187	alternative	evidence_to_proof_route_onramp.base.connect_proof.inference	Do not emit ProofConnectionGraph: Demote the unrestricted reduction route because its assertion license does not cross the model boundary.	rejected
falsifier.e7f4f7e7488b21	falsifier	evidence_to_proof_route_onramp.base.connect_proof.inference	A counterexample to the gadget preservation lemma closes the route.	Not recorded
alternative.f30ea382dda95e	alternative	evidence_to_proof_route_onramp.base.evaluate_hypotheses.inference	Do not emit HypothesisAssessment: Demote the unrestricted reduction route because its assertion license does not cross the model boundary.	accepted
falsifier.5c45d27958e537	falsifier	evidence_to_proof_route_onramp.base.evaluate_hypotheses.inference	A counterexample to the gadget preservation lemma closes the route.	Not recorded
alternatives_and_failure.244a2560566cdfe	failure	evidence_to_proof_route_onramp.base.evaluate_hypotheses.inference	evaluate_hypotheses rejects HypothesisAssessment: Demote the unrestricted reduction route because its assertion license does not cross the model boundary.	Not recorded
alternative.7c0ea84c932fdb	alternative	evidence_to_proof_route_onramp.base.audit_hubs.inference	Do not emit HubBiasAudit: Demote the unrestricted reduction route because its assertion license does not cross the model boundary.	rejected

ID	Kind	Inference	Description	Disposition
falsifier.a3dd870b693e8f	falsifier	evidence_to_proof_route_onramp.base.audit_hubs.inference	A counterexample to the gadget preservation lemma closes the route.	Not recorded
alternative.549c52fdf9427f	alternative	evidence_to_proof_route_onramp.base.find_entrypoint.inference	Do not emit EntrypointCandidates: Demote the unrestricted reduction route because its assertion license does not cross the model boundary.	rejected
falsifier.8a984ea6224bf6	falsifier	evidence_to_proof_route_onramp.base.find_entrypoint.inference	A counterexample to the gadget preservation lemma closes the route.	Not recorded
alternative.718960964b4131	alternative	evidence_to_proof_route_onramp.base.pathfind.inference	Do not emit ProofRouteCandidates: Demote the unrestricted reduction route because its assertion license does not cross the model boundary.	rejected
falsifier.c826e7307f06d9	falsifier	evidence_to_proof_route_onramp.base.pathfind.inference	A counterexample to the gadget preservation lemma closes the route.	Not recorded
alternative.2e09561a019cad	alternative	evidence_to_proof_route_onramp.base.audit_regime_shift.inference	Do not emit RegimeShiftAudit: Demote the unrestricted reduction route because its assertion license does not cross the model boundary.	rejected
falsifier.d38138a2cfdcaf	falsifier	evidence_to_proof_route_onramp.base.audit_regime_shift.inference	A counterexample to the gadget preservation lemma closes the route.	Not recorded
alternative.5edb2627508cba	alternative	evidence_to_proof_route_onramp.base.hunt_counterexamples.inference	Do not emit FalsificationPacket: Demote the unrestricted reduction route because its assertion license does not cross the model boundary.	rejected
falsifier.3fa965347548d8	falsifier	evidence_to_proof_route_onramp.base.hunt_counterexamples.inference	A counterexample to the gadget preservation lemma closes the route.	Not recorded
alternative.0a4d095e562c3e	alternative	evidence_to_proof_route_onramp.base.expert_handoff.inference	Do not emit ExpertRouteDossier: Demote the unrestricted reduction route because its assertion license does not cross the model boundary.	rejected
falsifier.9f065afc3ae566	falsifier	evidence_to_proof_route_onramp.base.expert_handoff.inference	A counterexample to the gadget preservation lemma closes the route.	Not recorded

7 Cross-cutting validation ledgers

7.1 Scope transitions

Semantic ID	Complete recorded detail
evidence_to_proof_route_onramp.base.ingest.inference	{"event_id": "evidence_to_proof_route_onramp.base.ingest.inference", "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "step_id": "ingest"}
evidence_to_proof_route_onramp.base.parse_pvsnp.inference	{"event_id": "evidence_to_proof_route_onramp.base.parse_pvsnp.inference", "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "step_id": "parse_pvsnp"}
evidence_to_proof_route_onramp.base.normalize_theorems.inference	{"event_id": "evidence_to_proof_route_onramp.base.normalize_theorems.inference", "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "step_id": "normalize_theorems"}
evidence_to_proof_route_onramp.base.ground_conjecture.inference	{"event_id": "evidence_to_proof_route_onramp.base.ground_conjecture.inference", "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "step_id": "ground_conjecture"}
evidence_to_proof_route_onramp.base.narrate_literature_evolution.inference	{"event_id": "evidence_to_proof_route_onramp.base.narrate_literature_evolution.inference", "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "step_id": "narrate_literature_evolution"}
evidence_to_proof_route_onramp.base.connect_proof.inference	{"event_id": "evidence_to_proof_route_onramp.base.connect_proof.inference", "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "step_id": "connect_proof"}
evidence_to_proof_route_onramp.base.evaluate_hypotheses.inference	{"event_id": "evidence_to_proof_route_onramp.base.evaluate_hypotheses.inference", "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "step_id": "evaluate_hypotheses"}
evidence_to_proof_route_onramp.base.audit_hubs.inference	{"event_id": "evidence_to_proof_route_onramp.base.audit_hubs.inference", "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "step_id": "audit_hubs"}
evidence_to_proof_route_onramp.base.find_entrypoint.inference	{"event_id": "evidence_to_proof_route_onramp.base.find_entrypoint.inference", "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "step_id": "find_entrypoint"}
evidence_to_proof_route_onramp.base.pathfind.inference	{"event_id": "evidence_to_proof_route_onramp.base.pathfind.inference", "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "step_id": "pathfind"}
evidence_to_proof_route_onramp.base.audit_regime_shift.inference	{"event_id": "evidence_to_proof_route_onramp.base.audit_regime_shift.inference", "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "step_id": "audit_regime_shift"}
evidence_to_proof_route_onramp.base.hunt_counterexamples.inference	{"event_id": "evidence_to_proof_route_onramp.base.hunt_counterexamples.inference", "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "step_id": "hunt_counterexamples"}
evidence_to_proof_route_onramp.base.expert_handoff.inference	{"event_id": "evidence_to_proof_route_onramp.base.expert_handoff.inference", "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "step_id": "expert_handoff"}

7.2 Authority boundaries

Semantic ID	Complete recorded detail
evidence_to_proof_route_onramp.base.ingest.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "evidence_to_proof_route_onramp.base.ingest.inference", "exercised": ["ingest-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "ingest"}
evidence_to_proof_route_onramp.base.parse_pvsnp.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "evidence_to_proof_route_onramp.base.parse_pvsnp.inference", "exercised": ["parse_pvsnp-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "parse_pvsnp"}
evidence_to_proof_route_onramp.base.normalize_theorems.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "evidence_to_proof_route_onramp.base.normalize_theorems.inference", "exercised": ["normalize_theorems-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "normalize_theorems"}
evidence_to_proof_route_onramp.base.ground_conjecture.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "evidence_to_proof_route_onramp.base.ground_conjecture.inference", "exercised": ["ground_conjecture-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "ground_conjecture"}
evidence_to_proof_route_onramp.base.narrate_literature_evolution.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "evidence_to_proof_route_onramp.base.narrate_literature_evolution.inference", "exercised": ["narrate_literature_evolution-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "narrate_literature_evolution"}
evidence_to_proof_route_onramp.base.connect_proof.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "evidence_to_proof_route_onramp.base.connect_proof.inference", "exercised": ["connect_proof-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "connect_proof"}
evidence_to_proof_route_onramp.base.evaluate_hypotheses.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "evidence_to_proof_route_onramp.base.evaluate_hypotheses.inference", "exercised": ["evaluate_hypotheses-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "evaluate_hypotheses"}
evidence_to_proof_route_onramp.base.audit_hubs.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "evidence_to_proof_route_onramp.base.audit_hubs.inference", "exercised": ["audit_hubs-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "audit_hubs"}
evidence_to_proof_route_onramp.base.find_entrypoint.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "evidence_to_proof_route_onramp.base.find_entrypoint.inference", "exercised": ["find_entrypoint-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "find_entrypoint"}
evidence_to_proof_route_onramp.base.pathfind.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "evidence_to_proof_route_onramp.base.pathfind.inference", "exercised": ["pathfind-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "pathfind"}
evidence_to_proof_route_onramp.base.audit_regime_shift.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "evidence_to_proof_route_onramp.base.audit_regime_shift.inference", "exercised": ["audit_regime_shift-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "audit_regime_shift"}
evidence_to_proof_route_onramp.base.hunt_counterexamples.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "evidence_to_proof_route_onramp.base.hunt_counterexamples.inference", "exercised": ["hunt_counterexamples-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "hunt_counterexamples"}
evidence_to_proof_route_onramp.base.expert_handoff.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "evidence_to_proof_route_onramp.base.expert_handoff.inference", "exercised": ["expert_handoff-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "expert_handoff"}

7.3 Open and discharged obligations

Semantic ID	Complete recorded detail
obligation.bd06003cd83102	{"event_id": "evidence_to_proof_route_onramp.base.evaluate_hypotheses.inference", "obligation_id": "obligation.bd06003cd83102", "requirement": "A counterexample to the gadget preservation lemma closes the route.", "status": "open"}

7.4 Handoffs

Semantic ID	Complete recorded detail
evidence_to_proof_route_onramp.handoff	{"accepted": true, "destination_workflow_id": "theorem_proof_delivery", "event_id": "evidence_to_proof_route_onramp.handoff"}

A Complete chronological lifecycle ledger

This ledger records execution order. A precedes relation is not evidential support or entailment.

Event ID	Seq.	Family	Step	Disposition	Parents
evidence_to_proof_route_onramp.base.ingest.control	1	authorization	ingest	authorized	None
evidence_to_proof_route_onramp.base.ingest.inference	2	inference	ingest	accepted	["evidence_to_proof_route_onramp.base.ingest.control"]
evidence_to_proof_route_onramp.base.parse_pvsnp.control	3	authorization	parse_pvsnp	authorized	["evidence_to_proof_route_onramp.base.ingest.inference"]
evidence_to_proof_route_onramp.base.parse_pvsnp.inference	4	inference	parse_pvsnp	accepted	["evidence_to_proof_route_onramp.base.parse_pvsnp.control"]
evidence_to_proof_route_onramp.base.normalize_theorems.control	5	authorization	normalize_theorems	authorized	["evidence_to_proof_route_onramp.base.parse_pvsnp.inference"]
evidence_to_proof_route_onramp.base.normalize_theorems.inference	6	inference	normalize_theorems	accepted	["evidence_to_proof_route_onramp.base.normalize_theorems.control"]
evidence_to_proof_route_onramp.base.ground_conjecture.control	7	authorization	ground_conjecture	authorized	["evidence_to_proof_route_onramp.base.normalize_theorems.inference"]
evidence_to_proof_route_onramp.base.ground_conjecture.inference	8	inference	ground_conjecture	accepted	["evidence_to_proof_route_onramp.base.ground_conjecture.control"]
evidence_to_proof_route_onramp.base.narrate_literature_evolution.control	9	authorization	narrate_literature_evolution	authorized	["evidence_to_proof_route_onramp.base.ground_conjecture.inference"]

Event ID	Seq.	Family	Step	Disposition	Parents
evidence_to_proof_route_onramp.base.narrate_literature_evolution.inference	10	inference	narrate_literature_evolution	accepted	["evidence_to_proof_route_onramp.base.narrate_literature_evolution.control"]
evidence_to_proof_route_onramp.base.connect_proof.control	11	authorization	connect_proof	authorized	["evidence_to_proof_route_onramp.base.narrate_literature_evolution.inference"]
evidence_to_proof_route_onramp.base.connect_proof.inference	12	inference	connect_proof	accepted	["evidence_to_proof_route_onramp.base.connect_proof.control"]
evidence_to_proof_route_onramp.base.evaluate_hypotheses.control	13	authorization	evaluate_hypotheses	authorized	["evidence_to_proof_route_onramp.base.connect_proof.inference"]
evidence_to_proof_route_onramp.base.evaluate_hypotheses.inference	14	failure	evaluate_hypotheses	failed	["evidence_to_proof_route_onramp.base.evaluate_hypotheses.control"]
evidence_to_proof_route_onramp.base.audit_hubs.control	15	authorization	audit_hubs	authorized	["evidence_to_proof_route_onramp.base.evaluate_hypotheses.inference"]
evidence_to_proof_route_onramp.base.audit_hubs.inference	16	inference	audit_hubs	accepted	["evidence_to_proof_route_onramp.base.audit_hubs.control"]
evidence_to_proof_route_onramp.base.find_entrypoint.control	17	authorization	find_entrypoint	authorized	["evidence_to_proof_route_onramp.base.audit_hubs.inference"]
evidence_to_proof_route_onramp.base.find_entrypoint.inference	18	inference	find_entrypoint	accepted	["evidence_to_proof_route_onramp.base.find_entrypoint.control"]
evidence_to_proof_route_onramp.base.pathfind.control	19	authorization	pathfind	authorized	["evidence_to_proof_route_onramp.base.find_entrypoint.inference"]
evidence_to_proof_route_onramp.base.pathfind.inference	20	inference	pathfind	accepted	["evidence_to_proof_route_onramp.base.pathfind.control"]
evidence_to_proof_route_onramp.base.audit_regime_shift.control	21	authorization	audit_regime_shift	authorized	["evidence_to_proof_route_onramp.base.pathfind.inference"]
evidence_to_proof_route_onramp.base.audit_regime_shift.inference	22	inference	audit_regime_shift	accepted	["evidence_to_proof_route_onramp.base.audit_regime_shift.control"]
evidence_to_proof_route_onramp.base.hunt_counterexamples.control	23	authorization	hunt_counterexamples	authorized	["evidence_to_proof_route_onramp.base.audit_regime_shift.inference"]

Event ID	Seq.	Family	Step	Disposition	Parents
evidence_to_proof_route_onramp.base.hunt_counterexamples.inference	24	inference	hunt_counterexamples	accepted	["evidence_to_proof_route_onramp.base.hunt_counterexamples.control"]
evidence_to_proof_route_onramp.base.expert_handoff.control	25	authorization	expert_handoff	authorized	["evidence_to_proof_route_onramp.base.hunt_counterexamples.inference"]
evidence_to_proof_route_onramp.base.expert_handoff.inference	26	inference	expert_handoff	accepted	["evidence_to_proof_route_onramp.base.expert_handoff.control"]
evidence_to_proof_route_onramp.handoff	27	handoff	Not recorded	handed_off	["evidence_to_proof_route_onramp.base.expert_handoff.inference"]

B Claim ledger

Semantic ID	Complete recorded detail
evidence_to_proof_route_onramp.base.ingest.inference	{"claim_id": "evidence_to_proof_route_onramp.base.ingest.inference", "epistemic_status": "evidence_backed", "event_id": "evidence_to_proof_route_onramp.base.ingest.inference", "evidence_ids": ["evidence.b26c5b14983132"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "text": "ingest produces IngestedKnowledgeBase: Retain the local-reduction route only in the declared finite regime.", "trace_incomplete": false, "workflow_disposition": "accepted"}
evidence_to_proof_route_onramp.base.parse_pvsnp.inference	{"claim_id": "evidence_to_proof_route_onramp.base.parse_pvsnp.inference", "epistemic_status": "evidence_backed", "event_id": "evidence_to_proof_route_onramp.base.parse_pvsnp.inference", "evidence_ids": ["evidence.3232ddcaf03acc"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "text": "parse_pvsnp produces ParsedKnowledgeBase: Retain the local-reduction route only in the declared finite regime.", "trace_incomplete": false, "workflow_disposition": "accepted"}
evidence_to_proof_route_onramp.base.normalize_theorems.inference	{"claim_id": "evidence_to_proof_route_onramp.base.normalize_theorems.inference", "epistemic_status": "evidence_backed", "event_id": "evidence_to_proof_route_onramp.base.normalize_theorems.inference", "evidence_ids": ["evidence.c6621527158116"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "text": "normalize_theorems produces NormalizedTheoremCorpus: Retain the local-reduction route only in the declared finite regime.", "trace_incomplete": false, "workflow_disposition": "accepted"}
evidence_to_proof_route_onramp.base.ground_conjecture.inference	{"claim_id": "evidence_to_proof_route_onramp.base.ground_conjecture.inference", "epistemic_status": "evidence_backed", "event_id": "evidence_to_proof_route_onramp.base.ground_conjecture.inference", "evidence_ids": ["evidence.ff0cd87caa476a"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "text": "ground_conjecture produces GroundedConjecture: Retain the local-reduction route only in the declared finite regime.", "trace_incomplete": false, "workflow_disposition": "accepted"}

Semantic ID	Complete recorded detail
evidence_to_proof_route_onramp.base.narrate_literature_evolution.inference	{"claim_id": "evidence_to_proof_route_onramp.base.narrate_literature_evolution.inference", "epistemic_status": "evidence_backed", "event_id": "evidence_to_proof_route_onramp.base.narrate_literature_evolution.inference", "evidence_ids": ["evidence.6d9bdc9d2ecdd2"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "text": "narrate_literature_evolution produces TargetLiteratureNarrative: Retain the local-reduction route only in the declared finite regime.", "trace_incomplete": false, "workflow_disposition": "accepted"}
evidence_to_proof_route_onramp.base.connect_proof.inference	{"claim_id": "evidence_to_proof_route_onramp.base.connect_proof.inference", "epistemic_status": "evidence_backed", "event_id": "evidence_to_proof_route_onramp.base.connect_proof.inference", "evidence_ids": ["evidence.a5cf3aa8c367cd"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "text": "connect_proof produces ProofConnectionGraph: Retain the local-reduction route only in the declared finite regime.", "trace_incomplete": false, "workflow_disposition": "accepted"}
evidence_to_proof_route_onramp.base.evaluate_hypotheses.inference	{"claim_id": "evidence_to_proof_route_onramp.base.evaluate_hypotheses.inference", "epistemic_status": "quarantined", "event_id": "evidence_to_proof_route_onramp.base.evaluate_hypotheses.inference", "evidence_ids": ["evidence.c1929579a0c9b9"], "integrity_status": "integrity_valid", "obligation_ids": ["obligation.bd06003cd83102"], "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "text": "evaluate_hypotheses rejects HypothesisAssessment: Demote the unrestricted reduction route because its assertion license does not cross the model boundary.", "trace_incomplete": false, "workflow_disposition": "failed"}
evidence_to_proof_route_onramp.base.audit_hubs.inference	{"claim_id": "evidence_to_proof_route_onramp.base.audit_hubs.inference", "epistemic_status": "evidence_backed", "event_id": "evidence_to_proof_route_onramp.base.audit_hubs.inference", "evidence_ids": ["evidence.a1e8d5b2868a2b"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "text": "audit_hubs produces HubBiasAudit: Retain the local-reduction route only in the declared finite regime.", "trace_incomplete": false, "workflow_disposition": "accepted"}
evidence_to_proof_route_onramp.base.find_entrypoint.inference	{"claim_id": "evidence_to_proof_route_onramp.base.find_entrypoint.inference", "epistemic_status": "evidence_backed", "event_id": "evidence_to_proof_route_onramp.base.find_entrypoint.inference", "evidence_ids": ["evidence.cf493724160827"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "text": "find_entrypoint produces EntrypointCandidates: Retain the local-reduction route only in the declared finite regime.", "trace_incomplete": false, "workflow_disposition": "accepted"}
evidence_to_proof_route_onramp.base.pathfind.inference	{"claim_id": "evidence_to_proof_route_onramp.base.pathfind.inference", "epistemic_status": "evidence_backed", "event_id": "evidence_to_proof_route_onramp.base.pathfind.inference", "evidence_ids": ["evidence.acd4492512064a"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "text": "pathfind produces ProofRouteCandidates: Retain the local-reduction route only in the declared finite regime.", "trace_incomplete": false, "workflow_disposition": "accepted"}
evidence_to_proof_route_onramp.base.audit_regime_shift.inference	{"claim_id": "evidence_to_proof_route_onramp.base.audit_regime_shift.inference", "epistemic_status": "evidence_backed", "event_id": "evidence_to_proof_route_onramp.base.audit_regime_shift.inference", "evidence_ids": ["evidence.dff0e8ffaf0efe"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "text": "audit_regime_shift produces RegimeShiftAudit: Retain the local-reduction route only in the declared finite regime.", "trace_incomplete": false, "workflow_disposition": "accepted"}

Semantic ID	Complete recorded detail
evidence_to_proof_route_onramp.base.hunt_counterexamples.inference	{"claim_id": "evidence_to_proof_route_onramp.base.hunt_counterexamples.inference", "epistemic_status": "evidence_backed", "event_id": "evidence_to_proof_route_onramp.base.hunt_counterexamples.inference", "evidence_ids": ["evidence.a069114828c61a"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "text": "hunt_counterexamples produces FalsificationPacket: Retain the local-reduction route only in the declared finite regime.", "trace_incomplete": false, "workflow_disposition": "accepted"}
evidence_to_proof_route_onramp.base.expert_handoff.inference	{"claim_id": "evidence_to_proof_route_onramp.base.expert_handoff.inference", "epistemic_status": "evidence_backed", "event_id": "evidence_to_proof_route_onramp.base.expert_handoff.inference", "evidence_ids": ["evidence.099972813773cf"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "licensed proof route", "before": "source claim", "model": "local reductions", "regime": "finite"}, "text": "expert_handoff produces ExpertRouteDossier: Retain the local-reduction route only in the declared finite regime.", "trace_incomplete": false, "workflow_disposition": "accepted"}

C Typed evidence ledger

Semantic ID	Complete recorded detail
evidence.b26c5b14983132	{"artifact_type": "SourceBundle", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "evidence_to_proof_route_onramp.base.ingest.inference", "evidence_id": "evidence.b26c5b14983132", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.3232ddcaf03acc	{"artifact_type": "IngestedKnowledgeBase", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "evidence_to_proof_route_onramp.base.parse_pvsnp.inference", "evidence_id": "evidence.3232ddcaf03acc", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.c6621527158116	{"artifact_type": "IngestedKnowledgeBase", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "evidence_to_proof_route_onramp.base.normalize_theorems.inference", "evidence_id": "evidence.c6621527158116", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.ff0cd87caa476a	{"artifact_type": "NormalizedTheoremCorpus", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "evidence_to_proof_route_onramp.base.ground_conjecture.inference", "evidence_id": "evidence.ff0cd87caa476a", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.6d9bdc9d2ecdd2	{"artifact_type": "GroundedConjecture", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "evidence_to_proof_route_onramp.base.narrate_literature_evolution.inference", "evidence_id": "evidence.6d9bdc9d2ecdd2", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.a5cf3aa8c367cd	{"artifact_type": "GroundedConjecture", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "evidence_to_proof_route_onramp.base.connect_proof.inference", "evidence_id": "evidence.a5cf3aa8c367cd", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.c1929579a0c9b9	{"artifact_type": "ProofConnectionGraph", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "evidence_to_proof_route_onramp.base.evaluate_hypotheses.inference", "evidence_id": "evidence.c1929579a0c9b9", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "refutes"}

Semantic ID	Complete recorded detail
evidence.a1e8d5b2868a2b	{"artifact_type": "ProofConnectionGraph", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "evidence_to_proof_route_onramp.base.audit_hubs.inference", "evidence_id": "evidence.a1e8d5b2868a2b", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.cf493724160827	{"artifact_type": "ProofConnectionGraph", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "evidence_to_proof_route_onramp.base.find_entrypoint.inference", "evidence_id": "evidence.cf493724160827", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.acd4492512064a	{"artifact_type": "EntrypointCandidates", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "evidence_to_proof_route_onramp.base.pathfind.inference", "evidence_id": "evidence.acd4492512064a", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.dff0e8ffaf0efe	{"artifact_type": "ProofRouteCandidates", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "evidence_to_proof_route_onramp.base.audit_regime_shift.inference", "evidence_id": "evidence.dff0e8ffaf0efe", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.a069114828c61a	{"artifact_type": "ProofRouteCandidates", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "evidence_to_proof_route_onramp.base.hunt_counterexamples.inference", "evidence_id": "evidence.a069114828c61a", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.099972813773cf	{"artifact_type": "ProofRouteCandidates", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "evidence_to_proof_route_onramp.base.expert_handoff.inference", "evidence_id": "evidence.099972813773cf", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}

D Decision and gate ledger

Semantic ID	Complete recorded detail
evidence_to_proof_route_onramp.base.ingest.control	{"disposition": "authorized", "event_id": "evidence_to_proof_route_onramp.base.ingest.control", "family": "authorization", "reason": "", "step_id": "ingest"}
evidence_to_proof_route_onramp.base.parse_pvsnp.control	{"disposition": "authorized", "event_id": "evidence_to_proof_route_onramp.base.parse_pvsnp.control", "family": "authorization", "reason": "", "step_id": "parse_pvsnp"}
evidence_to_proof_route_onramp.base.normalize_theorems.control	{"disposition": "authorized", "event_id": "evidence_to_proof_route_onramp.base.normalize_theorems.control", "family": "authorization", "reason": "", "step_id": "normalize_theorems"}
evidence_to_proof_route_onramp.base.ground_conjecture.control	{"disposition": "authorized", "event_id": "evidence_to_proof_route_onramp.base.ground_conjecture.control", "family": "authorization", "reason": "", "step_id": "ground_conjecture"}
evidence_to_proof_route_onramp.base.narrate_literature_evolution.control	{"disposition": "authorized", "event_id": "evidence_to_proof_route_onramp.base.narrate_literature_evolution.control", "family": "authorization", "reason": "", "step_id": "narrate_literature_evolution"}
evidence_to_proof_route_onramp.base.connect_proof.control	{"disposition": "authorized", "event_id": "evidence_to_proof_route_onramp.base.connect_proof.control", "family": "authorization", "reason": "", "step_id": "connect_proof"}

Semantic ID	Complete recorded detail
evidence_to_proof_route_onramp.base.evaluate_hypotheses.control	{"disposition": "authorized", "event_id": "evidence_to_proof_route_onramp.base.evaluate_hypotheses.control", "family": "authorization", "reason": "", "step_id": "evaluate_hypotheses"}
evidence_to_proof_route_onramp.base.audit_hubs.control	{"disposition": "authorized", "event_id": "evidence_to_proof_route_onramp.base.audit_hubs.control", "family": "authorization", "reason": "", "step_id": "audit_hubs"}
evidence_to_proof_route_onramp.base.find_entrypoint.control	{"disposition": "authorized", "event_id": "evidence_to_proof_route_onramp.base.find_entrypoint.control", "family": "authorization", "reason": "", "step_id": "find_entrypoint"}
evidence_to_proof_route_onramp.base.pathfind.control	{"disposition": "authorized", "event_id": "evidence_to_proof_route_onramp.base.pathfind.control", "family": "authorization", "reason": "", "step_id": "pathfind"}
evidence_to_proof_route_onramp.base.audit_regime_shift.control	{"disposition": "authorized", "event_id": "evidence_to_proof_route_onramp.base.audit_regime_shift.control", "family": "authorization", "reason": "", "step_id": "audit_regime_shift"}
evidence_to_proof_route_onramp.base.hunt_counterexamples.control	{"disposition": "authorized", "event_id": "evidence_to_proof_route_onramp.base.hunt_counterexamples.control", "family": "authorization", "reason": "", "step_id": "hunt_counterexamples"}
evidence_to_proof_route_onramp.base.expert_handoff.control	{"disposition": "authorized", "event_id": "evidence_to_proof_route_onramp.base.expert_handoff.control", "family": "authorization", "reason": "", "step_id": "expert_handoff"}

E Artifact and receipt ledger

Semantic ID	Complete recorded detail
No artifact and receipt ledger recorded.	

F Specialized projection ledger

Semantic ID	Complete recorded detail
No specialized projection ledger recorded.	

G Causal and workflow graph ledger

Typed edge summary

precedes: 38, contains: 27, supports: 12, refutes: 1, produces: 1

Edge meanings are explicit: precedes is chronology; supports/refutes/constrains are epistemic relations; contains/produces/branches/joins/refines/validates/blocks/hands-off are workflow relations.

G.1 Nodes

Node	Type	Label	Status
evidence_to_proof_route_onramp.base.ingest.control	event	authorization	Not recorded
evidence_to_proof_route_onramp.base.ingest.inference	event	ingest produces IngestedKnowledgeBase: Retain the local-reduction route only in the declared finite regime.	Not recorded
evidence.b26c5b14983132	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
evidence_to_proof_route_onramp.base.parse_pvsnp.control	event	authorization	Not recorded
evidence_to_proof_route_onramp.base.parse_pvsnp.inference	event	parse_pvsnp produces ParsedKnowledgeBase: Retain the local-reduction route only in the declared finite regime.	Not recorded
evidence.3232ddcaf03acc	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
evidence_to_proof_route_onramp.base.normalize_theorems.control	event	authorization	Not recorded
evidence_to_proof_route_onramp.base.normalize_theorems.inference	event	normalize_theorems produces NormalizedTheoremCorpus: Retain the local-reduction route only in the declared finite regime.	Not recorded
evidence.c6621527158116	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
evidence_to_proof_route_onramp.base.ground_conjecture.control	event	authorization	Not recorded
evidence_to_proof_route_onramp.base.ground_conjecture.inference	event	ground_conjecture produces GroundedConjecture: Retain the local-reduction route only in the declared finite regime.	Not recorded
evidence.ff0cd87caa476a	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
evidence_to_proof_route_onramp.base.narrate_literature_evolution.control	event	authorization	Not recorded
evidence_to_proof_route_onramp.base.narrate_literature_evolution.inference	event	narrate_literature_evolution produces TargetLiteratureNarrative: Retain the local-reduction route only in the declared finite regime.	Not recorded
evidence.6d9bdc9d2ecdd2	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
evidence_to_proof_route_onramp.base.connect_proof.control	event	authorization	Not recorded
evidence_to_proof_route_onramp.base.connect_proof.inference	event	connect_proof produces ProofConnectionGraph: Retain the local-reduction route only in the declared finite regime.	Not recorded
evidence.a5cf3aa8c367cd	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
evidence_to_proof_route_onramp.base.evaluate_hypotheses.control	event	authorization	Not recorded
evidence_to_proof_route_onramp.base.evaluate_hypotheses.inference	event	evaluate_hypotheses rejects HypothesisAssessment: Demote the unrestricted reduction route because its assertion license does not cross the model boundary.	Not recorded
evidence.c1929579a0c9b9	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded

Node	Type	Label	Status
obligation.bd06003cd83102	obligation	A counterexample to the gadget preservation lemma closes the route.	Not recorded
evidence_to_proof_route_onramp.base.audit_hubs.control	event	authorization	Not recorded
evidence_to_proof_route_onramp.base.audit_hubs.inference	event	audit_hubs produces HubBiasAudit: Retain the local-reduction route only in the declared finite regime.	Not recorded
evidence.a1e8d5b2868a2b	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
evidence_to_proof_route_onramp.base.find_entrypoint.control	event	authorization	Not recorded
evidence_to_proof_route_onramp.base.find_entrypoint.inference	event	find_entrypoint produces Entrypoint Candidates: Retain the local-reduction route only in the declared finite regime.	Not recorded
evidence.cf493724160827	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
evidence_to_proof_route_onramp.base.pathfind.control	event	authorization	Not recorded
evidence_to_proof_route_onramp.base.pathfind.inference	event	pathfind produces ProofRouteCandidates: Retain the local-reduction route only in the declared finite regime.	Not recorded
evidence.acd4492512064a	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
evidence_to_proof_route_onramp.base.audit_regime_shift.control	event	authorization	Not recorded
evidence_to_proof_route_onramp.base.audit_regime_shift.inference	event	audit_regime_shift produces RegimeShifAudit: Retain the local-reduction route only in the declared finite regime.	Not recorded
evidence.dff0e8ffaf0efe	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
evidence_to_proof_route_onramp.base.hunt_counterexamples.control	event	authorization	Not recorded
evidence_to_proof_route_onramp.base.hunt_counterexamples.inference	event	hunt_counterexamples produces FalsificationPacket: Retain the local-reduction route only in the declared finite regime.	Not recorded
evidence.a069114828c61a	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
evidence_to_proof_route_onramp.base.expert_handoff.control	event	authorization	Not recorded
evidence_to_proof_route_onramp.base.expert_handoff.inference	event	expert_handoff produces ExpertRouteDossier: Retain the local-reduction route only in the declared finite regime.	Not recorded
evidence.099972813773cf	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
evidence_to_proof_route_onramp.handoff	event	Typed handoff to theorem_proof_delivery	Not recorded
step:ingest	step	ingest	observed
step:parse_pvsnp	step	parse_pvsnp	observed
step:normalize_theorems	step	normalize_theorems	observed
step:ground_conjecture	step	ground_conjecture	observed
step:narrate_literature_evolution	step	narrate_literature_evolution	observed
step:connect_proof	step	connect_proof	observed
step:evaluate_hypotheses	step	evaluate_hypotheses	observed

Node	Type	Label	Status
step:audit_hubs	step	audit_hubs	observed
step:find_entrypoint	step	find_entrypoint	observed
step:pathfind	step	pathfind	observed
step:audit_regime_shift	step	audit_regime_shift	observed
step:hunt_counterexamples	step	hunt_counterexamples	observed
step:expert_handoff	step	expert_handoff	observed

G.2 Complete typed edge table

Source	Relation	Target	Adapter / detail
evidence_to_proof_route_onramp.base.ingest.control	precedes	evidence_to_proof_route_onramp.base.ingest.inference	None
evidence.b26c5b14983132	supports	evidence_to_proof_route_onramp.base.ingest.inference	None
evidence_to_proof_route_onramp.base.ingest.inference	precedes	evidence_to_proof_route_onramp.base.parse_pvsnp.control	None
evidence_to_proof_route_onramp.base.parse_pvsnp.control	precedes	evidence_to_proof_route_onramp.base.parse_pvsnp.inference	None
evidence.3232ddcaf03acc	supports	evidence_to_proof_route_onramp.base.parse_pvsnp.inference	None
evidence_to_proof_route_onramp.base.parse_pvsnp.inference	precedes	evidence_to_proof_route_onramp.base.normalize_theorems.control	None
evidence_to_proof_route_onramp.base.normalize_theorems.control	precedes	evidence_to_proof_route_onramp.base.normalize_theorems.inference	None
evidence.c6621527158116	supports	evidence_to_proof_route_onramp.base.normalize_theorems.inference	None
evidence_to_proof_route_onramp.base.normalize_theorems.inference	precedes	evidence_to_proof_route_onramp.base.ground_conjecture.control	None
evidence_to_proof_route_onramp.base.ground_conjecture.control	precedes	evidence_to_proof_route_onramp.base.ground_conjecture.inference	None
evidence.ff0cd87caa476a	supports	evidence_to_proof_route_onramp.base.ground_conjecture.inference	None
evidence_to_proof_route_onramp.base.ground_conjecture.inference	precedes	evidence_to_proof_route_onramp.base.narrate_literature_evolution.control	None
evidence_to_proof_route_onramp.base.narrate_literature_evolution.control	precedes	evidence_to_proof_route_onramp.base.narrate_literature_evolution.inference	None
evidence.6d9bdc9d2ecdd2	supports	evidence_to_proof_route_onramp.base.narrate_literature_evolution.inference	None
evidence_to_proof_route_onramp.base.narrate_literature_evolution.inference	precedes	evidence_to_proof_route_onramp.base.connect_proof.control	None
evidence_to_proof_route_onramp.base.connect_proof.control	precedes	evidence_to_proof_route_onramp.base.connect_proof.inference	None

Source	Relation	Target	Adapter / detail
evidence.a5cf3aa8c367cd	supports	evidence_to_proof_route_onramp.base.connect_proof.inference	None
evidence_to_proof_route_onramp.base.connect_proof.inference	precedes	evidence_to_proof_route_onramp.base.evaluate_hypotheses.control	None
evidence_to_proof_route_onramp.base.evaluate_hypotheses.control	precedes	evidence_to_proof_route_onramp.base.evaluate_hypotheses.inference	None
evidence.c1929579a0c9b9	refutes	evidence_to_proof_route_onramp.base.evaluate_hypotheses.inference	None
evidence_to_proof_route_onramp.base.evaluate_hypotheses.inference	produces	obligation.bd06003cd83102	None
evidence_to_proof_route_onramp.base.evaluate_hypotheses.inference	precedes	evidence_to_proof_route_onramp.base.audit_hubs.control	None
evidence_to_proof_route_onramp.base.audit_hubs.control	precedes	evidence_to_proof_route_onramp.base.audit_hubs.inference	None
evidence.a1e8d5b2868a2b	supports	evidence_to_proof_route_onramp.base.audit_hubs.inference	None
evidence_to_proof_route_onramp.base.audit_hubs.inference	precedes	evidence_to_proof_route_onramp.base.find_entrypoint.control	None
evidence_to_proof_route_onramp.base.find_entrypoint.control	precedes	evidence_to_proof_route_onramp.base.find_entrypoint.inference	None
evidence.cf493724160827	supports	evidence_to_proof_route_onramp.base.find_entrypoint.inference	None
evidence_to_proof_route_onramp.base.find_entrypoint.inference	precedes	evidence_to_proof_route_onramp.base.pathfind.control	None
evidence_to_proof_route_onramp.base.pathfind.control	precedes	evidence_to_proof_route_onramp.base.pathfind.inference	None
evidence.acd4492512064a	supports	evidence_to_proof_route_onramp.base.pathfind.inference	None
evidence_to_proof_route_onramp.base.pathfind.inference	precedes	evidence_to_proof_route_onramp.base.audit_regime_shift.control	None
evidence_to_proof_route_onramp.base.audit_regime_shift.control	precedes	evidence_to_proof_route_onramp.base.audit_regime_shift.inference	None
evidence.dff0e8ffaf0efe	supports	evidence_to_proof_route_onramp.base.audit_regime_shift.inference	None
evidence_to_proof_route_onramp.base.audit_regime_shift.inference	precedes	evidence_to_proof_route_onramp.base.hunt_counterexamples.control	None
evidence_to_proof_route_onramp.base.hunt_counterexamples.control	precedes	evidence_to_proof_route_onramp.base.hunt_counterexamples.inference	None
evidence.a069114828c61a	supports	evidence_to_proof_route_onramp.base.hunt_counterexamples.inference	None
evidence_to_proof_route_onramp.base.hunt_counterexamples.inference	precedes	evidence_to_proof_route_onramp.base.expert_handoff.control	None

Source	Relation	Target	Adapter / detail
evidence_to_proof_route_onramp.base.expert_handoff.control	precedes	evidence_to_proof_route_onramp.base.expert_handoff.inference	None
evidence.099972813773cf	supports	evidence_to_proof_route_onramp.base.expert_handoff.inference	None
evidence_to_proof_route_onramp.base.expert_handoff.inference	precedes	evidence_to_proof_route_onramp.handoff	None
step:ingest	contains	evidence_to_proof_route_onramp.base.ingest.control	None
step:ingest	contains	evidence_to_proof_route_onramp.base.ingest.inference	None
step:parse_pvsnp	contains	evidence_to_proof_route_onramp.base.parse_pvsnp.control	None
step:parse_pvsnp	contains	evidence_to_proof_route_onramp.base.parse_pvsnp.inference	None
step:normalize_theorems	contains	evidence_to_proof_route_onramp.base.normalize_theorems.control	None
step:normalize_theorems	contains	evidence_to_proof_route_onramp.base.normalize_theorems.inference	None
step:ground_conjecture	contains	evidence_to_proof_route_onramp.base.ground_conjecture.control	None
step:ground_conjecture	contains	evidence_to_proof_route_onramp.base.ground_conjecture.inference	None
step:narrate_literature_evolution	contains	evidence_to_proof_route_onramp.base.narrate_literature_evolution.control	None
step:narrate_literature_evolution	contains	evidence_to_proof_route_onramp.base.narrate_literature_evolution.inference	None
step:connect_proof	contains	evidence_to_proof_route_onramp.base.connect_proof.control	None
step:connect_proof	contains	evidence_to_proof_route_onramp.base.connect_proof.inference	None
step:evaluate_hypotheses	contains	evidence_to_proof_route_onramp.base.evaluate_hypotheses.control	None
step:evaluate_hypotheses	contains	evidence_to_proof_route_onramp.base.evaluate_hypotheses.inference	None
step:evaluate_hypotheses	contains	evidence_to_proof_route_onramp.base.evaluate_hypotheses.inference	None
step:audit_hubs	contains	evidence_to_proof_route_onramp.base.audit_hubs.control	None
step:audit_hubs	contains	evidence_to_proof_route_onramp.base.audit_hubs.inference	None
step:find_entrypoint	contains	evidence_to_proof_route_onramp.base.find_entrypoint.control	None
step:find_entrypoint	contains	evidence_to_proof_route_onramp.base.find_entrypoint.inference	None

Source	Relation	Target	Adapter / detail
step:pathfind	contains	evidence_to_proof_route_onramp.base.pathfind.control	None
step:pathfind	contains	evidence_to_proof_route_onramp.base.pathfind.inference	None
step:audit_regime_shift	contains	evidence_to_proof_route_onramp.base.audit_regime_shift.control	None
step:audit_regime_shift	contains	evidence_to_proof_route_onramp.base.audit_regime_shift.inference	None
step:hunt_counterexamples	contains	evidence_to_proof_route_onramp.base.hunt_counterexamples.control	None
step:hunt_counterexamples	contains	evidence_to_proof_route_onramp.base.hunt_counterexamples.inference	None
step:expert_handoff	contains	evidence_to_proof_route_onramp.base.expert_handoff.control	None
step:expert_handoff	contains	evidence_to_proof_route_onramp.base.expert_handoff.inference	None
step:ingest	precedes	step:parse_pvsnp	None
step:parse_pvsnp	precedes	step:normalize_theorems	None
step:normalize_theorems	precedes	step:ground_conjecture	None
step:ground_conjecture	precedes	step:narrate_literature_evolution	None
step:narrate_literature_evolution	precedes	step:connect_proof	None
step:connect_proof	precedes	step:evaluate_hypotheses	None
step:evaluate_hypotheses	precedes	step:audit_hubs	None
step:audit_hubs	precedes	step:find_entrypoint	None
step:find_entrypoint	precedes	step:pathfind	None
step:pathfind	precedes	step:audit_regime_shift	None
step:audit_regime_shift	precedes	step:hunt_counterexamples	None
step:hunt_counterexamples	precedes	step:expert_handoff	None

H Projection completeness and omission ledger

Packet field	Disposition	Destination	Reason
alternatives_and_failures	rendered	negative-path body	Not recorded
artifacts_and_receipts	summarized_with_complete_appendix	artifact appendix	Not recorded
authority_ledger	rendered	authority body and appendix	Not recorded
claims	summarized_with_complete_appendix	claim ledger appendix	Not recorded
coverage	rendered	validation dashboard	Not recorded
decisions_and_gates	summarized_with_complete_appendix	decision and gate appendix	Not recorded
declared_route	rendered	workflow route body	Not recorded
evidence	summarized_with_complete_appendix	evidence ledger appendix	Not recorded
graph	summarized_with_complete_appendix	graph summary and complete edge appendix	Not recorded
handoffs	rendered	handoff body	Not recorded
identity	rendered	cover identity	Not recorded
inferences	rendered	expert inference dossiers	Not recorded

Packet field	Disposition	Destination	Reason
lens	rendered	cover and print profile	Not recorded
obligations	rendered	obligation body and appendix	Not recorded
profile_topology	rendered	route topology body	Not recorded
schema_version	rendered	build provenance	Not recorded
scope_ledger	rendered	scope body and appendix	Not recorded
specialized_projections	summarized_with_complete_appendix	specialized projection appendix	Not recorded
timeline	summarized_with_complete_appendix	complete chronology appendix	Not recorded

I Build provenance

Source packet lens

{"accent": "#55c2ff", "focus_question": "How did source evidence become a falsification-aware proof route?", "labels": {"claims": "Grounded claims", "failures": "Bias, regime, and falsification checks", "handoffs": "Expert route dossier"}, "lens_id": "evidence_to_proof_route_onramp", "mandatory_spine": ["ingest", "normalize_theorems", "ground_conjecture", "connect_proof", "audit_hubs", "find_entrypoint", "pathfind", "hunt_counterexamples", "expert_handoff"], "metrics": ["semantic_coverage", "evidence_resolution", "open_obligations", "trace_gaps"], "required_views": ["route", "inferences", "claims", "failures", "scope", "authority", "handoffs", "coverage"], "spine_label": "Provenance to expert route", "visible_name": "Evidence to Proof Route Onramp"}

Packet SHA-256

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Renderer version

1.0

Print inventory schema

1.0